

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Acrylamide : Bis-Acrylamide 37.5:1 (40% Solution/Electrophoresis)

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>BP1410-1</b>                                                                             |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                      |             |
|------------------------------------------------------|-------------|
| Acute Oral Toxicity                                  | Category 3  |
| Acute Inhalation Toxicity - Vapors                   | Category 4  |
| Skin Corrosion/Irritation                            | Category 2  |
| Serious Eye Damage/Eye Irritation                    | Category 2  |
| Skin Sensitization                                   | Category 1  |
| Germ Cell Mutagenicity                               | Category 1B |
| Carcinogenicity                                      | Category 1B |
| Reproductive Toxicity                                | Category 2  |
| Specific target organ toxicity - (repeated exposure) | Category 1  |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

#### Hazard Statements

H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H340 - May cause genetic defects  
H350 - May cause cancer  
H361 - Suspected of damaging fertility or the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H301 - Toxic if swallowed  
H332 - Harmful if inhaled

#### Precautionary Statements

##### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

##### Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

##### Storage

P403 - Store in a well-ventilated place

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

#### Other hazards which do not result in classification

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component              | CAS No    | Weight % |
|------------------------|-----------|----------|
| Water                  | 7732-18-5 | 60       |
| Acrylamide             | 79-06-1   | 39       |
| Methylene diacrylamide | 110-26-9  | 1        |

## Section 4 - First Aid Measures

#### Description of first aid measures

##### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                     |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Eye Contact</b>                         | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing                                                                                                        |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                         |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Ammonia.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

#### Environmental Precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

#### Methods for Containment and Clean Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Keep in a dry place. Keep container tightly closed. Protect from direct sunlight. Store under an inert atmosphere. To maintain product quality: Keep refrigerated.

#### Incompatible Materials

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component  | New Zealand WEL                       | Australia                   | ACGIH TLV                           | The United Kingdom                                                                     |
|------------|---------------------------------------|-----------------------------|-------------------------------------|----------------------------------------------------------------------------------------|
| Acrylamide | TWA: 0.0015 mg/m <sup>3</sup><br>Skin | TWA: 0.03 mg/m <sup>3</sup> | TWA: 0.03 mg/m <sup>3</sup><br>Skin | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

**Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene, Natural rubber, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Clear                    |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | 6.3                      |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Acrylamide                                     | -1.24                    |                                          |
| Methylene diacrylamide                         | -1.52                    |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |

|                          |                         |             |
|--------------------------|-------------------------|-------------|
| Vapor Density            | No data available       | (Air = 1.0) |
| Particle characteristics | Not applicable (liquid) |             |

Other information

## Section 10 - Stability and Reactivity

|                                  |                                                            |
|----------------------------------|------------------------------------------------------------|
| Reactivity                       | None known, based on information available                 |
| Stability                        | Air sensitive. Light sensitive.                            |
| Sensitivity to Mechanical Impact | No information available                                   |
| Sensitivity to Static Discharge  | No information available                                   |
| Hazardous Polymerization         | No information available.                                  |
| Hazardous Reactions              | None under normal processing.                              |
| Conditions to Avoid              | Exposure to air, Exposure to light, Incompatible products. |
| Incompatible Materials           | Strong oxidizing agents.                                   |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Ammonia.

## Section 11 - Toxicological Information

Acute Effects

Information on likely routes of exposure

**Product Information**

|            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | May produce an allergic reaction.                                                                                                        |
| Eyes       | Avoid contact with eyes. Irritating to eyes. Sensitization.                                                                              |
| Skin       | Avoid contact with skin. May cause irritation. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. |
| Ingestion  | May cause allergic reaction. May be harmful if swallowed.                                                                                |

Numerical measures of toxicity

**(a) acute toxicity;**

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

Toxicology data for the components

| Component              | LD50 Oral            | LD50 Dermal           | LC50 Inhalation |
|------------------------|----------------------|-----------------------|-----------------|
| Water                  | -                    | -                     | -               |
| Acrylamide             | 124 mg/kg ( Rat )    | 1141 mg/kg ( Rabbit ) |                 |
| Methylene diacrylamide | 50-300 mg/kg ( Rat ) | 1141 mg/kg (Rabbit)   |                 |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** Category 1

**Sensitization** May cause sensitization by skin contact

**(e) germ cell mutagenicity;** Category 1B

Mutagenic

**(f) carcinogenicity;** Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen  
Possible cancer hazard. May cause cancer based on animal data

| Component  | New Zealand          | Australia | New South Wales | Western Australia | IARC     | EU           | UK | Germany |
|------------|----------------------|-----------|-----------------|-------------------|----------|--------------|----|---------|
| Acrylamide | Confirmed carcinogen |           |                 |                   | Group 2A | Carc Cat. 1B |    | Cat. 2  |

**(g) reproductive toxicity;**  
**Reproductive Effects**

Category 2  
Experiments have shown reproductive toxicity effects on laboratory animals

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** Category 1

**Target Organs** Peripheral Nervous System (PNS).

**(j) aspiration hazard;** No data available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed**

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity**

Do not empty into drains. Contains a substance which is: Harmful to aquatic organisms.  
The product contains following substances which are hazardous for the environment.

| Component  | Freshwater Fish                                                                                                          | Water Flea                                                                                      | Freshwater Algae | Microtox |
|------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------|----------|
| Acrylamide | 124 mg/L LC50 96 h<br>74-150 mg/L LC50 96 h<br>81-150 mg/L LC50 96 h<br>103-115 mg/L LC50 96 h<br>137-191 mg/L LC50 96 h | EC50: = 98 mg/L, 48h<br>Flow through (Daphnia magna)<br>EC50: = 98 mg/L, 48h<br>(Daphnia magna) |                  |          |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Expected to be biodegradable

**Degradation in sewage treatment** Contains substances known to be hazardous to the environment or not degradable in waste

plant water treatment plants.

**Bioaccumulative Potential** No information available

| Component              | log Pow | Bioconcentration factor (BCF) |
|------------------------|---------|-------------------------------|
| Acrylamide             | -1.24   | No data available             |
| Methylene diacrylamide | -1.52   | No data available             |

**Mobility** .

**Other adverse effects**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## Section 14 - Transport Information

| Component                    | Hazchem Code |
|------------------------------|--------------|
| Acrylamide<br>79-06-1 ( 39 ) | 2X           |

**NZS 5433:2020** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known



## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

| Component  | New Zealand          |
|------------|----------------------|
| Acrylamide | Confirmed carcinogen |

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acrylamide | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 29. (see link for restriction details)<br>Use restricted. See item 60. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - 201-173-7 - Carcinogenic, Article 57a; Mutagenic, Article 57b                   |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component  | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Water      | 7732-18-5 | X     | X    | 231-791-2 | -      | -   | KE-35400 | X     | X    |
| Acrylamide | 79-06-1   | X     | X    | 201-173-7 | -      | -   | KE-29374 | X     | X    |

| Methylene diacrylamide | 110-26-9  | X    | X                                             | 203-750-9 | -   | -    | KE-23800 | X    | X    |
|------------------------|-----------|------|-----------------------------------------------|-----------|-----|------|----------|------|------|
|                        |           |      |                                               |           |     |      |          |      |      |
| Component              | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive |           | DSL | NDSL | PICCS    | ISHL | ENCS |
| Water                  | 7732-18-5 | X    | ACTIVE                                        |           | X   | -    | X        | -    | X    |
| Acrylamide             | 79-06-1   | X    | ACTIVE                                        |           | X   | -    | X        | X    | X    |
| Methvlene diacrylamide | 110-26-9  | X    | ACTIVE                                        |           | X   | -    | X        | X    | X    |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

NZIoC - New Zealand Inventory of Chemicals

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZS 5433:2020 - Transport of Dangerous Goods on Land

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

AICS - Australian Inventory of Chemical Substances

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists

PNEC - Predicted No Effect Concentration

OECD - Organisation for Economic Co-operation and Development

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Revision Date 10-Mar-2023

Revision Summary Not applicable

### Disclaimer

---

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**